NCAN · WADSWORTH · SUMMER 2026

Does cortex track the reflex as the spinal cord learns?

Machine learning on 20-year-old paired EMG + ECoG recordings.
Six animals, 2.7 million trials.

Suchith Rao & Tarun Senthil • with Dr. Carp • 21 August 2026

BACKGROUND

The paradigm, and the open question

What the lab does

Rats are rewarded for making the soleus H-reflex smaller (down-conditioning) or larger (up). Over weeks the spinal cord itself changes. Success is a ≥20% shift.

The open question

Boulay et al. showed cortical activity relates to reflex size. Does that relationship CHANGE as the animal learns — and can we see it on single trials?

Why machine learning helps here

Rather than hand-picking features from the cortical trace, we let a neural network learn its own compact description of each trial, then test what that description can predict. With 2.7 million trials, that is tractable in a way manual feature engineering is not.

METHODS

The pipeline, end to end

1	**Raw 2006 database**	MyISAM tables from the Elizan III system, loaded into a local MySQL server.
2	**Decode each trial**	Little-endian int16, two channels: soleus EMG and cortical ECoG. 150 samples at 5 kHz = 30 ms.
3	**Measure the physiology**	M-wave (2–4 ms), H-reflex (6–9 ms), and pre-stimulus background — all background-subtracted.
4	**Store as Zarr**	One array store per animal: signals, labels, timestamps, covariates. ~2.7M trials total.
5	**Learn a representation**	Autoencoder compresses each cortical trace to 48 numbers, trained across animals with no labels.
6	**Test what it predicts**	Freeze it, then ask what the representation says about the reflex — with controls.

Everything is scripted and reproducible; one command per animal.

METHODS · MEASUREMENT

Getting the reflex measurement right

Raw H is not interpretable	Reflex size scales with stimulus strength. Our M-wave drifted 100→450 µV over weeks, which alone made an early result look backwards.
Control for stimulus	The M-wave is the direct muscle response in the same trace — a per-trial receipt for how much stimulus arrived.
Control for excitability	Background EMG in the 20 ms before the stimulus. High background means an excited motoneuron pool and a larger H — not learning.
Subtract background	We use mean rectified amplitude, and rectified background does not cancel; left in, it pulls H/M toward 1 and can hide a real effect.

Every number in this talk is corrected for stimulus and background.

RESULTS · DATASET

Six animals, both directions

6	**2.7M**	**3 / 3**	**45–210**
animals	trials	down / up	days each

Finding good up-conditioned animals without downloading everything

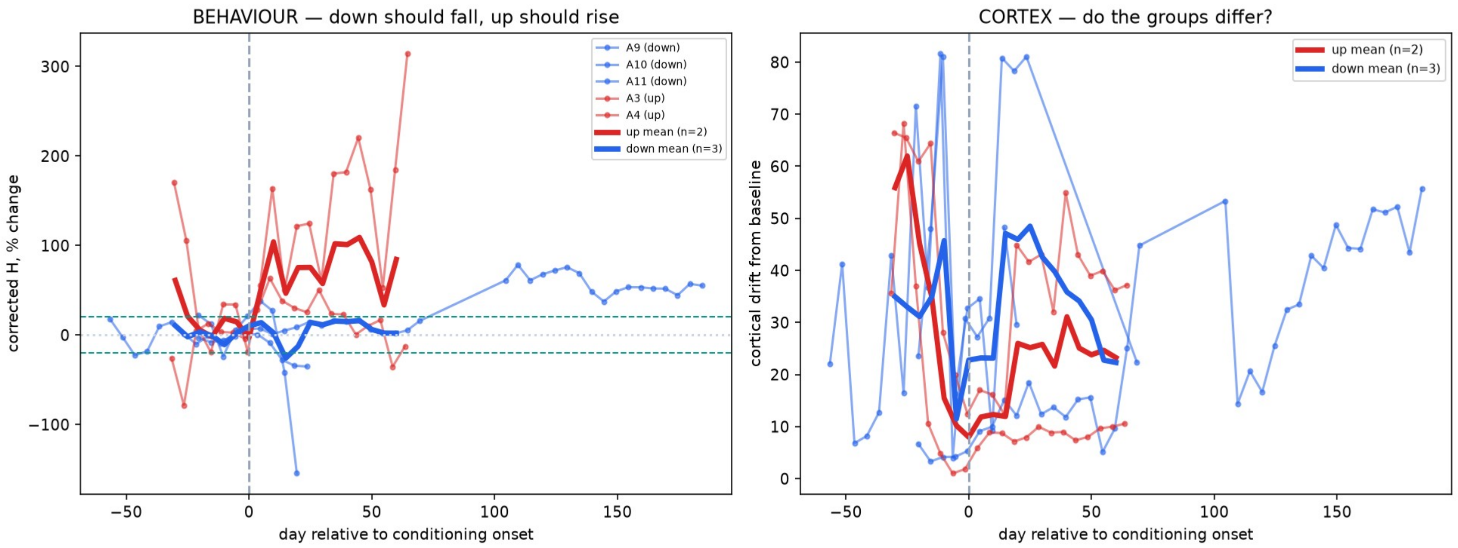
The log files are tiny. The reward criterion is a success readout — the experimenter raises the bar when the rat beats it. Animal 4's went 150→220, animal 3's 90→190. Animals 1 and 6 had theirs lowered, so they failed. We only downloaded the ones that worked.

One animal excluded

Animal 12's pre-stimulus cortical signal is 2.8 µV, versus 106–128 µV in the others — the ECoG electrode was effectively dead. Its apparent weak conditioning was a recording failure.

RESULT 1 · BEHAVIOUR

Conditioning works, in both directions



The positive control

Each animal aligned to its own training start. Red (up) rises, blue (down) falls — a 72 percentage-point group separation.

Three of five animals clear the lab's 20% criterion in the trained direction.

The two that don't become internal controls later in the talk.

RESULT 2 · A NEGATIVE RESULT

Average cortical state drifts — but not from learning

Our first approach was to ask whether the average cortical state moves across conditioning. It does. But three controls say that movement is the recording changing over weeks, not learning.

Baseline sham	Split baseline-only data as a fake experiment. Nothing happened, so it should be flat.	**Drifts as much as the real thing — 5 of 6 animals**
Randomised order	Shuffle which trial belongs to which day. Breaks time structure, keeps the numbers.	**Null ≈ 0 — so the metric itself is sound**
Direction contrast	Electrode drift is direction-independent; learning is not.	**Up 21.9 vs down 34.6, $t = -1.01$, n.s.**

In a single animal, learning and electrode drift are both smooth in time — they cannot be separated. That is a design limit, not a data problem.

APPROACH

A question that drift cannot contaminate

Instead of

"Did the average cortical state move over weeks?"

We ask

"Can cortex predict this trial's reflex size?"

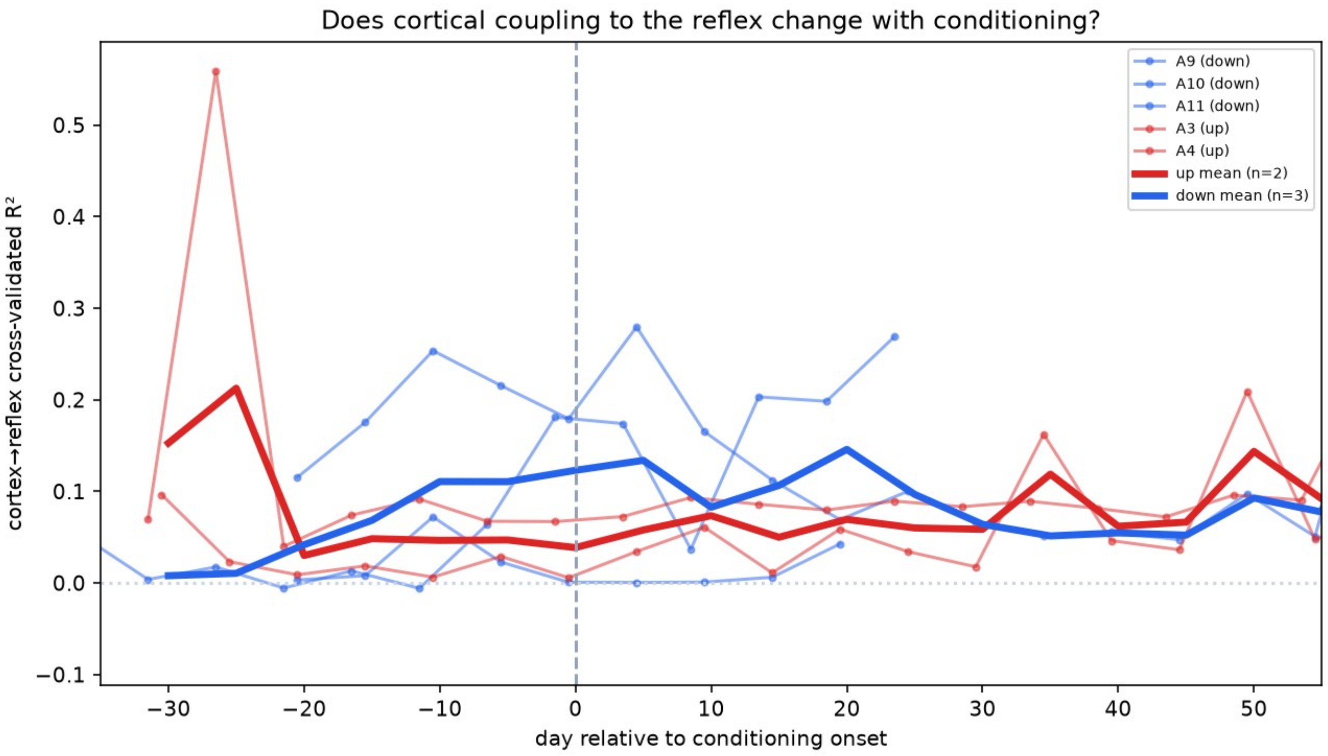
Why drift cannot fake this

The decoder is trained and tested inside a single 5-day block, by cross-validation. A slow electrode change shifts the training and test trials together, so it cancels. Drift can move a mean; it cannot create trial-by-trial predictive structure within a block.

- Stimulus and background removed inside each block
- Compared against a shuffle of the same block's labels

RESULT 3 · THE POSITIVE FINDING

Cortex predicts reflex size on single trials



Present in every animal

Cross-validated R^2 is above the shuffle null in every animal and nearly every block, after stimulus and background are removed.

Conservative estimate: $R^2 \approx 0.04$
null ≈ −0.01

Small, but consistent and not explainable by stimulus, muscle tone, or drift.

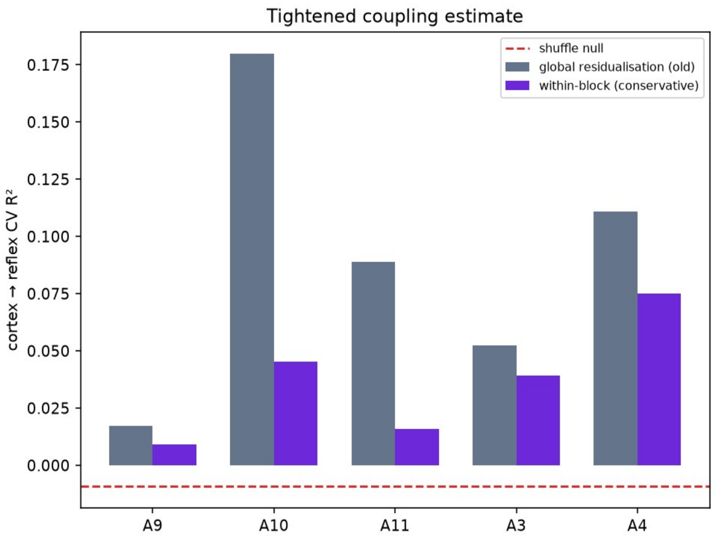
RESULT 3 · IN CONTEXT

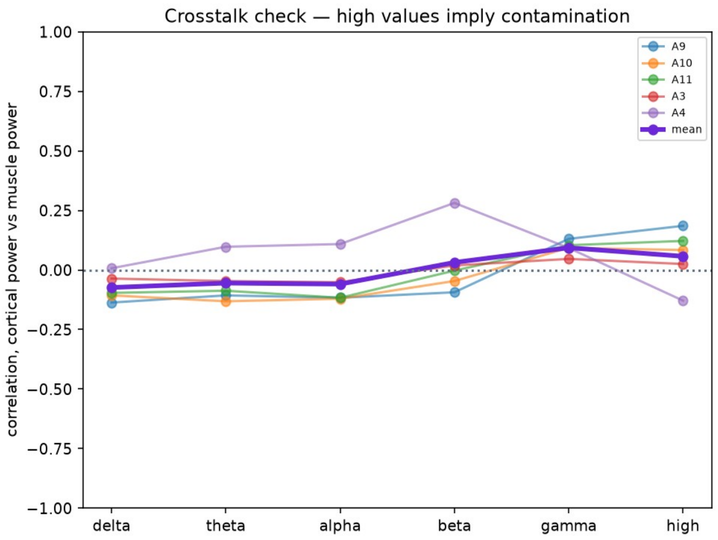
What actually explains reflex size?

Stimulus (M-wave)	47%	By far the dominant factor — how much stimulus reached the muscle.
Background excitability	1%	Small on its own, but a necessary control.
Cortical activity, alone	21%	Mostly overlapping with stimulus, since stimulus drives both.
Cortex, unique contribution	2%	What cortex adds once stimulus and background are already known.
Unexplained	51%	Trial-to-trial variability we cannot yet account for.

Percentages are variance in H-reflex amplitude explained, cross-validated within blocks, averaged over five animals.

Ruling out muscle contamination





The objection

Both channels record at once, so the ECoG electrode might simply be picking up muscle activity.

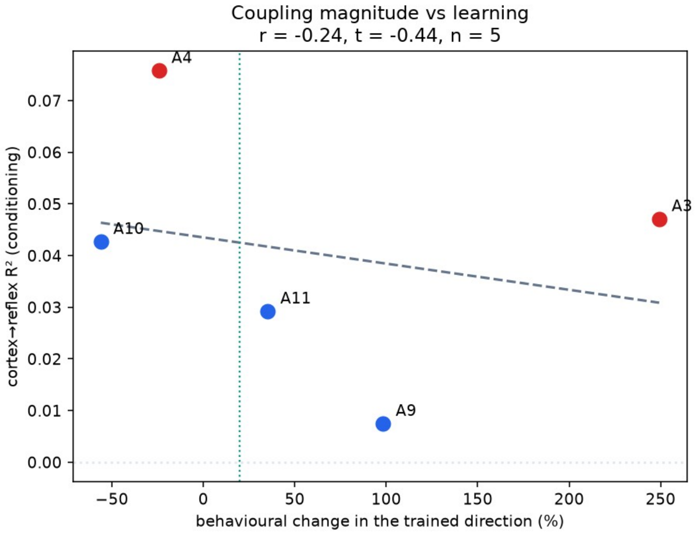
The test

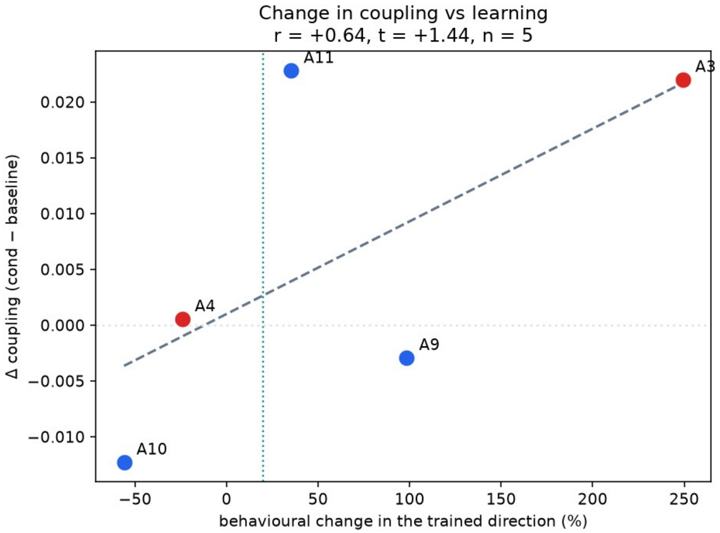
Correlate cortical against muscle power, band by band. Muscle artefact lives at high frequency, so contamination would show up there.

Above 100 Hz: r = +0.06.
No contamination.

RESULT 5 · THE INTERNAL CONTROL

Does the effect scale with how much they learned?





The logic

Animals differ in how much they learned, and two failed outright. If the cortical effect is tied to conditioning, it should track learning — and be absent in the animals that failed.

r = +0.64 between learning and the change in coupling.

Encouraging direction, but n = 5 — not significant. This is the analysis more animals would power.

SUMMARY

Where this stands

SOLID

Bidirectional conditioning

Reproduced with full stimulus and background control. Three of five animals clear the 20% criterion.

REAL, SMALL

Single-trial cortical coupling

$R^2 \approx 0.04$ over a ≈ 0 null, in every animal. Drift-immune by design; contamination ruled out.

NO

Average-state drift = learning

Three independent controls agree it reflects recording change. Worth reporting as a caution.

NEXT

What would strengthen this

1	**Add the remaining animals**	7, 8, 13, 14, 15 are scanned and ready. The lab aims for ~10 per group; we have 5 usable.
2	**Power the scaling test**	With ~10 animals the learning-vs-coupling correlation becomes a real test rather than a hint.
3	**Frequency-resolved coupling**	Which bands carry the reflex information? Sharpens both the result and the contamination argument.
4	**Time course**	Does cortical change lead or follow the behavioural change? Carp's suggestion, and mechanistically the interesting question.

Target venue: a machine-learning conference (ICML / NeurIPS), framed as an extension of Boulay et al.

QUESTIONS FOR YOU

Where we'd value your input

1 Is a 2% unique cortical contribution meaningful?

It is small but consistent and survives every control we can think of. Is that a result worth publishing?

2 Is the within-block design convincing?

Train and test inside one 5-day block so drift cancels. Does that satisfy you that the effect isn't recording change?

3 What else could produce this?

We've ruled out stimulus, muscle tone, drift, and crosstalk. What are we missing?

4 Is the negative result worth reporting?

Three controls say average-state drift reflects recording, not learning. Useful caution, or a distraction?

Code, figures and per-animal results: github.com/switchrao777/hroc-ncan